

AEDHunter: Investigating and Enhancing AED Retrieval through a Mobile Application

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Abstract

Early defibrillation improves survival rates in out-of-hospital cardiac arrest (OHCA), yet public use of automated external defibrillators (AEDs) remains low due to unawareness of their locations. Existing solutions like 2D maps are often insufficient during emergencies, especially indoors where navigation is challenging. We developed *AEDHunter*, a gamified, location-based mobile application that enhances public awareness of AED locations by transforming the retrieval process into an engaging adventure. *AEDHunter* leverages smartphone sensors to analyze user retrieval patterns and navigation behaviors in real-world environments. Through experiments involving repeated interactions with *AEDHunter*, we investigate how users' confidence, willingness, and behaviors in locating AEDs evolve. We also introduce a semi-supervised classification method to detect mobile device usage during these tasks, laying the groundwork for analyzing nuanced AED retrieval behaviors. This work-in-progress provides initial insights into enhancing AED awareness and retrieval through gamified applications.

CCS Concepts

• **Human-centered computing** → **Empirical studies in ubiquitous and mobile computing**; Empirical studies in HCI.

Keywords

Out-of-Hospital Cardiac Arrest, Mobile Sensing, AED Retrieval

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1 Introduction

Out-of-hospital cardiac arrest (OHCA) remains a major public health concern, with survival rates persistently low worldwide [17]. Early defibrillation can significantly increase survival chances; however, each minute of delay decreases survival by 7–10% [19, 24]. Although Automated External Defibrillators (AEDs) are strategically installed in many public areas, their usage during emergencies is alarmingly low. For instance, public-access AEDs are utilized in only 0.7% of OHCA cases in South Korea and 1.4% in Germany [20, 26], revealing a significant gap in public familiarity and awareness of AED locations.

Research indicates that public unawareness of AED locations is a significant barrier to their effective use [31, 37]. Surveys reveal that only a small fraction of non-professionals know the nearest AED location [4, 6, 18], and even among medical students, awareness is surprisingly low [34]. People tend to overlook AEDs in their everyday environments, highlighting the necessity for interventions that increase public attention to and familiarity with AED locations.

Gamification offers an avenue for enhancing engagement and learning across various domains [16, 36]. However, designing effective gamified solutions to improve AED retrieval requires foundational insight into real-world retrieval patterns. Insights into how individuals locate AEDs, the obstacles they encounter, and their decision-making processes can inform the development of gamification strategies that address these specific issues.

By examining retrieval patterns and user feedback, our work provides crucial groundwork for more robust gamified interventions aimed at improving emergency preparedness. The key contributions of this study are:

- We created AEDHunter, a mobile application that transforms locating AEDs into a simple exploration game to enhance public knowledge of AED locations.
- AEDHunter utilizes smartphone sensors, including GPS, Wi-Fi, IMU, barometer, and compass, to gather detailed data on user behaviors. This data enables in-depth understanding of retrieval patterns.
- Our analysis of collected data reveals that repeated use of AEDHunter increases users' efficiency, confidence, and willingness to retrieve AEDs.
- We introduced a semi-supervised autoencoder model to detect mobile device usage during AED retrieval tasks, with

limited labeled data. It can quantitatively verify the survey results and enable large-scale applications.

2 Related Work

Gamified AED Crowdsourcing. Gamification, which incorporates game-like elements into non-gaming contexts to increase user engagement [9, 29], is widely used in mobile applications for health and education [15, 36]. While few studies have applied gamification to AED awareness and maintenance, some initiatives have demonstrated its potential. For instance, Merchant et al. [22] incentivized participants with points for creating, voting on, and sharing educational designs to boost AED visibility. Other projects developed mobile applications and websites using gamification techniques like points and competitions to educate users about AED locations and motivate them to inspect device conditions [2, 10, 21].

Behavior Tracking Technologies. Mobile phones, equipped with powerful and affordable sensors like accelerometers, Wi-Fi, Bluetooth, and directional sensors, enable detailed tracking of environmental dynamics and human behavior. For instance, accelerometer data can identify daily activities (e.g., walking or driving) [1, 35], while combining it with GPS reveals transportation modes [1, 33]. Bluetooth and Wi-Fi data further aid in determining proximity, trajectories, and even room-level positioning [5, 7, 11, 14, 25, 32].

3 Motivations and Research Questions

Despite efforts to raise public awareness of AEDs through gamification [2, 10, 21], significant gaps remain in understanding and improving AED retrieval during emergencies. Current applications often overlook mobile sensing technologies that can monitor and analyze user behaviors, providing critical insights into retrieval patterns and environmental challenges when locating AEDs. Additionally, there's a lack of analysis on how behavioral changes during AED retrieval impact on emergency preparedness.

To address these issues, we developed AEDHunter, a mobile application that encourages users to actively locate AEDs using mobile sensing technologies. Leveraging built-in smartphone sensors like GPS and Wi-Fi, AEDHunter monitors user retrieval patterns and provides feedback on navigation behaviors. Our approach aims to understand user behaviors and challenges in locating AEDs, assess the impact of active exploration on emergency preparedness, and explore methods for behavior detection using sensor data. Our research focuses on:

RQ1: How to utilize sensor data to understand users' behaviors during AED retrieval tasks in real-world environments?

RQ2: How does actively locating AEDs through our application impact users' emergency preparedness?

4 System Development

4.1 Key Features

AEDHunter is designed not only to raise AED awareness and improve retrieval efficiency, but also to collect detailed behavioral and attitudinal data during actual usage. Its key features include (1) an emergency scenario simulation that replicates real-life situations requiring rapid response, (2) a feature displaying elapsed time and estimated survival probability to underscore the importance of swift retrieval, (3) adaptive navigation assistance that supports

varied navigational aids and helps researchers assess user performance and spatial awareness, (4) sensor data integration to capture user behavior and detect changes over time, (5) in-app surveys to track subjective feedback, and (6) gamification elements to promote exploration and sustained engagement. By combining these features, AEDHunter aims to enhance AED access while providing a robust framework for understanding and improving user behavior in urgent contexts.

4.2 User Interface and Operations

Figure 1 shows the AEDHunter user interface, where the main screen summarizes user progress. The AED hunting procedure includes five steps: (1) users encounter a simulated emergency with an image of an unresponsive individual and the prompt, "Someone has collapsed and is unresponsive. Please get an AED immediately to save them," initiating the hunt by tapping "Let's get the AED now!"; (2) they receive information about the target AED's location, including building name and floor; (3) users receive real-time navigation assistance displaying elapsed time, estimated survival probability (calculated using an exponential decay model fitted to empirical data [13]), remaining distance, and their current location. Participants are divided into two groups: one group receives map-based assistance showing both current and target locations, while the other group does not receive map assistance; (4) to confirm the AED, users hold their smartphone near a Bluetooth beacon installed near the AED for three seconds; and (5) upon successful verification, they earn points and are prompted to complete a brief in-app survey about memory reliance (Q1), perceived ease (Q2), pre-visit familiarity (Q3), post-visit confidence (Q4), post-visit assistance willingness (Q5) (details are in A.1).

4.3 Application Implementation

AEDHunter was developed as an iOS application using Flutter (version 3.16.5), an open-source cross-platform UI development software. To collect hardware and software sensor data on iOS, we used AWARE Framework [23]. The hardware sensors include 2D locations, Wi-Fi BSSID, barometer, accelerometer, gyroscope, magnetometer, and Bluetooth. The 2D location, Wi-Fi BSSID, and barometer data are collected every second, and the motion sensor data (i.e., accelerometer, gyroscope, and compass direction) are collected 100 Hz. Screen touch events, device lock/unlock status, and application status information are collected by every event as software sensor data.

Collected sensor data are temporarily saved into the smartphone's local storage as a CSV file and uploaded to the connected Google Cloud Storage. Questionnaire responses, the target AED in the game session, and points are saved into the Cloud Firestore with dictionary format. Cloud Firestore also stores information about AED locations, including latitude, longitude, altitude, and place names. The application can refer to the latest AED information from Cloud Firestore. Map displays are implemented using OpenStreetMap via the Flutter plugin `flutter_osm_plugin`¹.

¹https://pub.dev/packages/flutter_osm_plugin

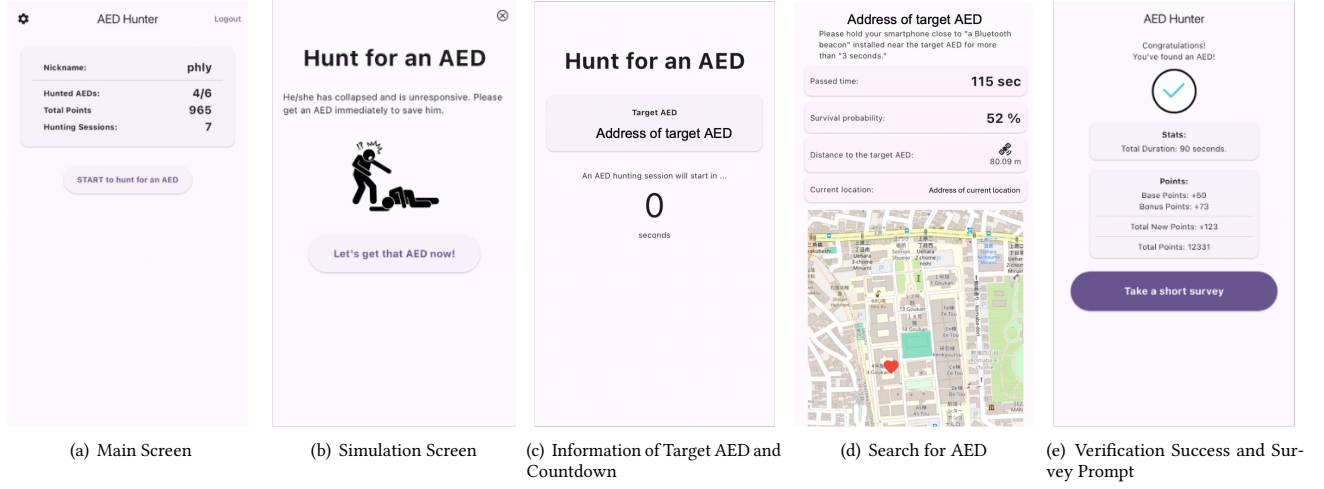


Figure 1: Screenshots of the AEDHunter

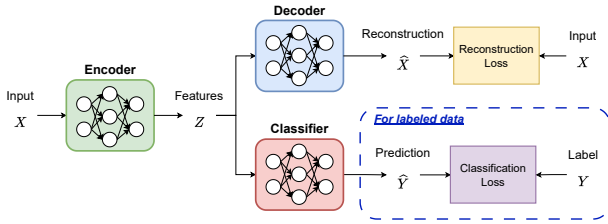


Figure 2: Architecture of semi-supervised autoencoder used for mobile phone usage detection during AED retrieval.

4.4 Detection of Mobile Phone Use During AED Retrieval

Monitoring mobile phone usage during AED retrieval provides insights into users' familiarity with AED locations and emergency preparedness. Manual data collection of phone usage is impractical for large-scale studies, necessitating automated detection methods. To address this, we employed a semi-supervised autoencoder model [12], leveraging abundant unlabeled data alongside limited labeled data [3]. As shown in Figure 2, our model comprises an encoder that compresses input features into a lower-dimensional representation, followed by a classifier to predict phone usage. A decoder reconstructs the original input [27], with reconstruction loss guiding the network to learn meaningful representations from both labeled and unlabeled data. The loss function is defined as:

$$L_i = \begin{cases} L_{\text{sup}}(\hat{y}_i, y_i) + \lambda L_{\text{rec}}(\hat{x}_i, x_i), & \text{if } i \text{ is labeled,} \\ \lambda L_{\text{rec}}(\hat{x}_i, x_i), & \text{if } i \text{ is unlabeled.} \end{cases}$$

where $L_{\text{sup}}(\hat{y}_i, y_i)$ represents the supervised loss between the predicted label $\hat{y}_i$ and the true label y_i , $L_{\text{rec}}(\hat{x}_i, x_i)$ denotes the reconstruction loss between the reconstructed input $\hat{x}_i$ and the original input x_i , and λ is a balancing hyperparameter.

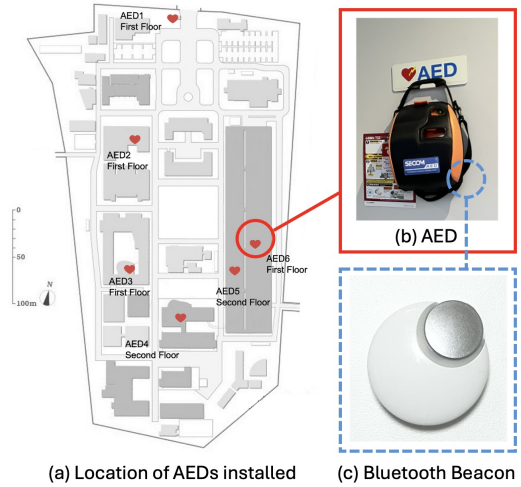


Figure 3: Experimental Environment. (a) Map showing AED locations (heart) and pedestrian pathways (grey lines). (b) A typical AED unit used in the study. (c) Example of Bluetooth Beacon placed near AEDs for arrival detection; each measures under 3.5 cm.

5 Experiments

We conducted real-world experiments to evaluate our system's effectiveness and study user behavior during interactions with it. The experiment took place on a university campus equipped with six AEDs, as shown in Figure 3.

Eight participants (5 males, 3 females; aged 22–32, average 25.4; iOS users), were recruited via social media. Before the experiment, participants completed a survey assessing their familiarity with the campus layout (Very Familiar, Familiar, Moderately Familiar, Limited Familiar, Minimally Familiar) and indicated any known

AED locations on campus. Participants varied in their knowledge of AED locations, from knowing exact spots to only general areas. To control for prior knowledge, participants were stratified into two groups of four, balancing familiarity levels: each group included one participant highly familiar with the campus, one with limited familiarity, and the remaining participants with moderate familiarity; knowledge of AED locations was evenly distributed between the groups. The Map Group received geo-map assistance to locate the AEDs during the experiment, while the No-Map Group did not receive any mapping assistance.

Participants engaged in game sessions at their convenience over a three-week period. To collect detailed behavioral data, we randomly selected eight game sessions for close observation, documenting participant actions as ground truth for training our mobile phone usage detection algorithms. Upon completion, participants filled out a post-experiment survey to provide feedback on the study.

For mobile phone use detection, we processed accelerometer and directional data to 20 Hz, using three-second intervals to predict phone usage during the central one-second window. Overlapping windows help capture motion patterns. We manually labeled 1,036 segments for phone usage, with 70% for training and 30% for evaluation, leaving 8,490 segments unlabeled. We used Mean Squared Error for reconstruction loss, Binary Cross Entropy for supervised loss, and set λ to 0.3. The encoder, decoder, and classifier were implemented with multilayer perceptrons (MLPs) [28] for simplicity and efficiency.

6 Results

6.1 Understanding Retrieval Behaviors through Sensor Data

Participants conducted 86 AED retrieval trips. The average retrieval time per trip was 134 seconds. The average distance between an AED and the user's starting location was 135.1 meters, calculated using the shortest path algorithm implemented in the NetworkX Python package [8]. Using GPS, accelerometer, WiFi, and iBeacon data, each visit was categorized into four phases, as shown in Figure 4: *Preparation* (average 3s), where users remained stationary or moved slowly, likely reviewing maps or planning routes; *Building Search* (average 75s), where participants walked along roads to locate buildings with the target AED; *Indoor AED Search* (average 54s), where users entered buildings to find the AED; and *Verifying* (average 3s), where users connected to an iBeacon to confirm the visit.

Our trajectory analysis revealed that participants using map assistance consistently chose shorter, more direct walking paths, minimizing time and distance. Initially, those without map assistance took more meandering routes and frequently backtracked, but with repeated exposure, they gradually adopted more efficient routes.

6.2 Impact of AEDHunter on Emergency Preparedness

We reviewed in-app survey responses collected over ten game sessions to evaluate how participants' views on AED retrieval behaviors changed over time. Both groups showed a positive trend in Q1–Q5 responses, indicating growing AED awareness, confidence, and willingness, as shown in Figure 5. Reliance on application guidance (Q1) decreased over time as participants developed rough ideas of AED locations. Those with map assistance found AEDs more easily from the start (Q2), while those without maps initially faced challenges but adapted quickly, with most rating the task as "Very Easy" by the tenth session. Familiarity with AED locations (Q3) increased across sessions, particularly among participants without map assistance, enhancing their environmental awareness. Confidence in retrieving AEDs (Q4) improved notably, especially among participants without map assistance, whose confidence stabilized at the maximum score after the seventh session. Willingness to assist others (Q5) remained high and increased slightly as familiarity grew.

Comparisons between initial and repeat visits to identical AED locations (Figure 6) highlight significant improvements during the second visit. Reliance on memory (Q1) increased substantially, and familiarity (Q3) was higher on repeat visits. Difficulty scores (Q2) decreased, while confidence (Q4) and willingness (Q5) to retrieve AEDs increased, indicating that repeated exposure significantly enhances AED location awareness and retrieval confidence.

6.3 Evaluation on Mobile Phone Usage Detection

We evaluated our semi-supervised method against a supervised baseline (without decoder and reconstruction loss) for detecting mobile phone usage during AED retrieval, using identical hyperparameters. As shown in Table 1, our method outperformed the baseline across all metrics, including precision, recall, and F1-score.

We applied the trained model to classify mobile device usage in each one-second window of unlabeled trips. Using the estimated mobile phone usage, we can calculate the proportion of time participants used their mobile devices for each trip. Specifically, Figure 7 illustrates the average proportion of time participants used their mobile devices during game sessions, categorized by their Q1 responses (scores of memory reliance). This trend demonstrates that memory reliance during AED retrieval is reflected in the proportion of time participants use their smartphones. The alignment between participants' self-reported reliance on memory and the model's detection of reduced mobile device usage supports the validity of our method. Furthermore, these results quantitatively verify that greater reliance on memory is associated with reduced mobile device usage during AED retrieval.

6.4 Identified Challenges in AED Retrieval

Despite AEDHunter's improvements in AED retrieval, challenges remain in indoor environments. Most participants using GPS navigation reported difficulties in accurately locating AEDs inside buildings. Limited GPS accuracy indoors, complex layouts, and multiple entrances often caused confusion, leading participants to choose the wrong entrances and increasing retrieval times. Trajectory data

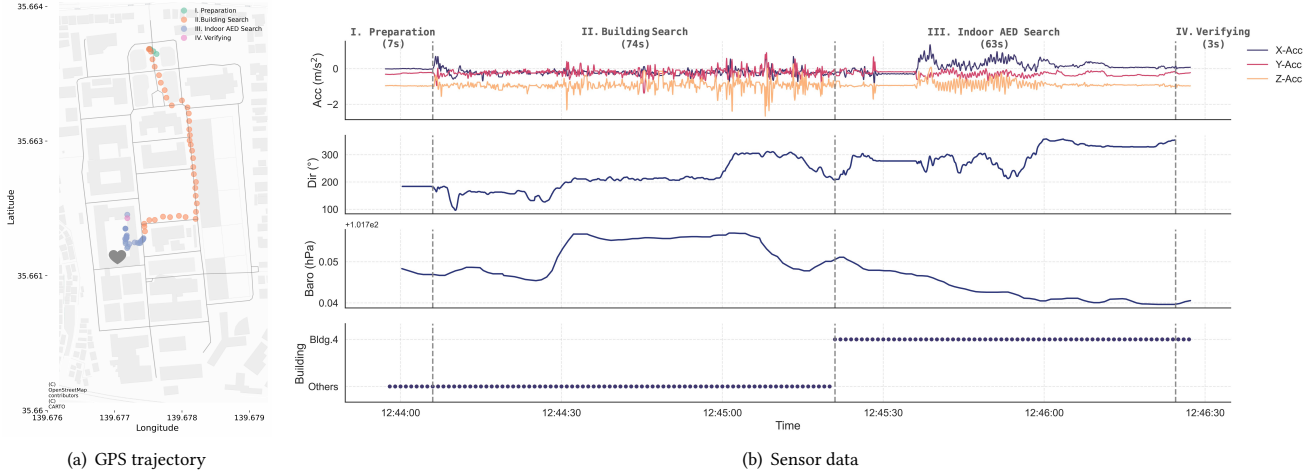


Figure 4: Example of collected data, including (a) a GPS trajectory that maps the participant’s movement path and (b) corresponding sensor data such as acceleration (Acc), direction (Dir), barometric pressure (Baro), and building information derived from WiFi BSSID. This data delineates the different phases of AED retrieval activity and the time allocated to each phase.

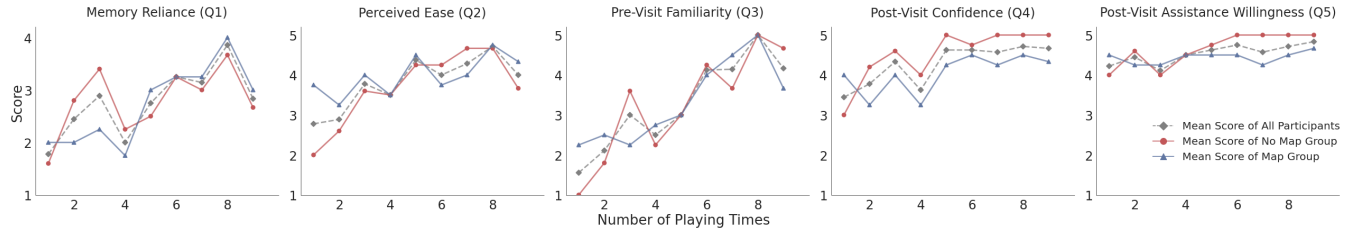


Figure 5: Scores of five questions in survey for two different groups

Method	Label	Precision	Recall	F1-score
Supervised	True	0.85	0.81	0.83
	False	0.70	0.76	0.73
Semi-supervised	True	0.90	0.87	0.88
	False	0.79	0.82	0.81

Table 1: Comparison of performance metrics between semi-supervised and supervised methods in mobile phone usage detection.

showed instances of backtracking and detours, highlighting the difficulties of indoor navigation even with map assistance. The details of the case study are available in Appendix A.2.

7 Discussion

Participants’ familiarity with AED locations improved significantly over multiple game sessions. Initially relying heavily on application guidance, they gradually developed destination (specific points), route (path sequences), and survey knowledge (overall area layout) [30]. This progression was more pronounced in participants without map assistance, indicating that experiential learning

through repeated tasks fosters deeper spatial understanding than reliance on navigation aids alone.

Participants sometimes faced significant difficulties during the Indoor AED Search phase due to limitations of current indoor navigation systems. Our gamified application mitigates these issues by encouraging users to explore and memorize AED locations and routes to AED. As familiarity increased, reliance on application guidance decreased, indicating a shift from technological dependence to spatial memory. This behavioral adaptation enhances efficient AED retrieval during emergencies, especially when technology may be unreliable.

Our mobile usage detection approach effectively leverages unlabeled data to achieve better performance than the supervised method (Table 1). This performance underscores the potential of our approach for practical applications where collecting labeled data is costly. Importantly, we found a strong correlation between higher reliance on memory and reduced mobile device usage (Figure 7). This suggests that monitoring device usage can measure reliance on navigation guidance.

Building on our findings, we have identified several implications for enhancing emergency response systems and gamified learning applications related to OHCA.

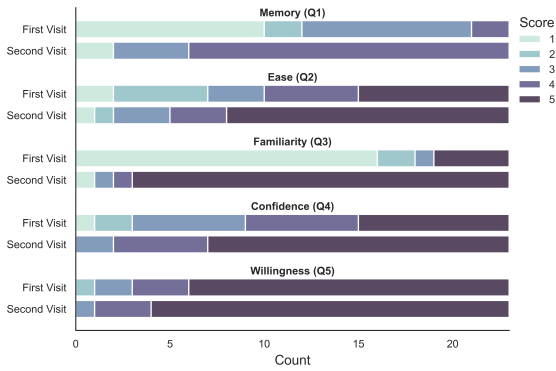


Figure 6: In-App survey results for first-time and second-time visits to the same AED locations.

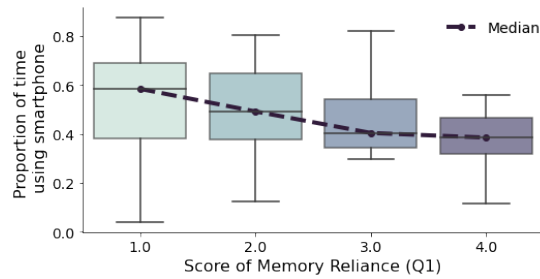


Figure 7: Correlation between mobile phone usage and memory reliance score; dashed lines connect median proportion per score group.

- **Promoting Spatial Knowledge Acquisition:** Incorporating features that further enhance route and survey knowledge—such as interactive maps, exploratory tasks, or augmented reality elements—can improve users’ ability to navigate complex environments efficiently. Encouraging exploration helps users develop mental maps, enhancing navigation in varied contexts. This aligns with our finding that experiential learning fosters deeper spatial understanding beyond mere destination knowledge.
- **Utilizing Behavioral Data for Adaptive Support:** Leveraging sensor data to monitor user interactions enables real-time assessment of navigational difficulties. Applications can adaptively provide additional support or guidance when increased device usage indicates potential challenges. This approach addresses the navigational obstacles identified in our study by offering timely assistance based on users’ behavior.
- **Enhancing Gamification Strategies:** Making the learning process more engaging and memorable through gamification motivates users to participate actively and retain critical spatial information. This approach is particularly effective in training scenarios aimed at improving emergency response capabilities. Our findings suggest that gamified experiential learning increases both competence and willingness to assist in emergencies.

8 Conclusion

In this study, we introduced *AEDHunter*, a mobile application that transforms locating AEDs into a simple exploration game. By investigating user retrieval patterns and interactions with nearby AEDs, as well as their feedback through in-app surveys, we verified that our application can enhance public awareness, increase confidence and willingness, and identify challenges in AED retrieval. Furthermore, we developed a semi-supervised method to detect mobile phone usage, which can quantitatively verify our survey results and enable broader applications. Overall, our study demonstrates the effectiveness of combining gamification with mobile sensing technologies to enhance AED awareness and retrieval behaviors. In the next step, we aim to expand the study by including more participants and environments, and to enhance the gamified elements for better user engagement and to promote spatial knowledge. Additionally, we plan to collect a more comprehensive WiFi BSSID map, which will allow us to conduct a deeper and more accurate analysis of indoor movements.

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References

- [1] Huthaifa I Ashqar, Mohammed H Almannaa, Mohammed Elhenawy, Hesham A Rakha, and Leanna House. 2018. Smartphone transportation mode recognition using a hierarchical machine learning classifier and pooled features from time and frequency domains. *IEEE Transactions on Intelligent Transportation Systems* 20, 1 (2018), 244–252.
- [2] Muhammad Imran Hakim Bin Hussein, Jun Hao Fong, Chu Xuan Lim, Jeannie SA Lee, Chek Tien Tan, and Yih Yng Ng. 2019. Mobile application for crowdsourced gamification of automated external defibrillator (AED) locations. In *Proceedings of the 4th international workshop on multimedia for personal health & health care*. 24–31.
- [3] Avrim Blum and Tom Mitchell. 1998. Combining labeled and unlabeled data with co-training. In *Proceedings of the eleventh annual conference on Computational learning theory*. 92–100.
- [4] Ben Brooks, Stephanie Chan, Peter Lander, Robbie Adamson, Gillian A Hodgetts, and Charles D Deakin. 2015. Public knowledge and confidence in the use of public access defibrillation. *Heart* 101, 12 (2015), 967–971.
- [5] E Cipriani, A Gemma, L Mannini, S Carrese, and U Crisalli. 2021. Traffic demand estimation using path information from Bluetooth data. *Transportation research part C: emerging technologies* 133 (2021), 103443.
- [6] KL Fan, LP Leung, HT Poon, HY Chiu, HL Liu, and WY Tang. 2016. Public knowledge of how to use an automatic external defibrillator in out-of-hospital cardiac arrest in Hong Kong. *Hong Kong Medical Journal* (2016).
- [7] Ramsey Faragher and Robert Harle. 2014. An analysis of the accuracy of bluetooth low energy for indoor positioning applications. In *Proceedings of the 27th International Technical Meeting of The Satellite Division of the Institute of Navigation (ION GNSS+ 2014)*. 201–210.
- [8] Aric Hagberg, Pieter J Swart, and Daniel A Schult. 2008. *Exploring network structure, dynamics, and function using NetworkX*. Technical Report. Los Alamos National Laboratory (LANL), Los Alamos, NM (United States).
- [9] Juho Hamari, Jonna Koivisto, and Harri Sarsa. 2014. Does gamification work?—a literature review of empirical studies on gamification. In *2014 47th Hawaii international conference on system sciences*. Ieee, 3025–3034.
- [10] Anran Hao, Chei Sian Lee, and Wesley Chee Wah Tan. 2019. Fun with AEDs: examining the effects of a gamified mobile crowdsourcing application. In *Proceedings of the 17th International Conference on Advances in Mobile Computing & Multimedia*. 207–211.
- [11] Raiful Hasan, Mohammad Aminul Hoque, Yasser Karim, Russell Griffin, David C Schwebel, and Ragib Hasan. 2022. Someone to watch over you: using Bluetooth beacons for alerting distracted pedestrians. *IEEE internet of things journal* 9, 22 (2022), 23017–23030.
- [12] Geoffrey E Hinton and Ruslan R Salakhutdinov. 2006. Reducing the dimensionality of data with neural networks. *science* 313, 5786 (2006), 504–507.
- [13] Mikael Holmberg, Stig Holmberg, and Johan Herlitz. 2000. Effect of bystander cardiopulmonary resuscitation in out-of-hospital cardiac arrest patients in Sweden.

- Resuscitation* 47, 1 (2000), 59–70.
- [14] Hande Hong, Chengwen Luo, and Mun Choon Chan. 2016. Socialprobe: Understanding social interaction through passive wifi monitoring. In *Proceedings of the 13th international conference on mobile and Ubiquitous systems: Computing, networking and services*. 94–103.
 - [15] Kai Huotari and Juho Hamari. 2012. Defining gamification: a service marketing perspective. In *Proceeding of the 16th international academic MindTrek conference*. 17–22.
 - [16] Dennis L Kappen and Lennart E Nacke. 2013. The kaleidoscope of effective gamification: deconstructing gamification in business applications. In *Proceedings of the first international conference on gameful design, research, and applications*. 119–122.
 - [17] Tekeyuki Kiguchi, Masashi Okubo, Chika Nishiyama, Ian Maconochie, Marcus Eng Hock Ong, Karl B Kern, Myra H Wyckoff, Bryan McNally, Erika F Christensen, Ingvald Tjelmeland, et al. 2020. Out-of-hospital cardiac arrest across the World: First report from the International Liaison Committee on Resuscitation (ILCOR). *Resuscitation* 152 (2020), 39–49.
 - [18] Dariusz Kozłowski, Tomasz Klosiewicz, Adam Kowalczyk, Anna Krystyna Kowalczyk, Edward Koźluk, Maria Dudziak, Wojciech Homenda, and Grzegorz Raczak. 2013. The knowledge of public access to defibrillation in selected cities in Poland. *Archives of Medical Science* 9, 1 (2013), 27–33.
 - [19] Mary P Larsen, Mickey S Eisenberg, Richard O Cummins, and Alfred P Hallstrom. 1993. Predicting survival from out-of-hospital cardiac arrest: a graphic model. *Annals of emergency medicine* 22, 11 (1993), 1652–1658.
 - [20] Jae Hyuk Lee, Dong Eun Lee, Hyun Wook Ryoo, Sungbae Moon, Jae Wan Cho, Yun Jeong Kim, Jong Kun Kim, Jung Ho Kim, Kyung Woo Lee, Sang-chan Jin, et al. 2021. Public awareness and willingness to use automated external defibrillators in a metropolitan city. *Clinical and Experimental Emergency Medicine* 8, 1 (2021), 1.
 - [21] Raina M Merchant, David A Asch, John C Hershey, Heather M Griffis, Shawndra Hill, Olivia Saynisch, Alison C Leung, Jeremy M Asch, Kirk Lozada, Lindsay D Nadkarni, et al. 2013. A crowdsourcing innovation challenge to locate and map automated external defibrillators. *Circulation: Cardiovascular Quality and Outcomes* 6, 2 (2013), 229–236.
 - [22] Raina M Merchant, Heather M Griffis, Yoonhee P Ha, Austin S Kilaru, Allison M Sellers, John C Hershey, Shawndra S Hill, Emily Kramer-Golinkoff, Lindsay Nadkarni, Margaret M Debski, et al. 2014. Hidden in plain sight: a crowdsourced public art contest to make automated external defibrillators more visible. *American journal of public health* 104, 12 (2014), 2306–2312.
 - [23] Yuuki Nishiyama, Denzil Ferreira, Yusaku Eigen, Wataru Sasaki, Tadashi Okoshi, Jin Nakazawa, Anind K Dey, and Kaoru Sezaki. 2020. IOS crowd-sensing won't hurt a bit!: AWARE framework and sustainable study guideline for iOS platform. In *Distributed, Ambient and Pervasive Interactions: 8th International Conference, DAPI 2020, Held as Part of the 22nd HCI International Conference, HCII 2020, Copenhagen, Denmark, July 19–24, 2020, Proceedings* 22. Springer, 223–243.
 - [24] Marcus Eng Hock Ong, Win Wah, Li Yang Hsu, Yih Ying Ng, Benjamin Sieu-Hon Leong, E Shaun Goh, Han Nee Gan, Lai Peng Tham, Rabind Antony Charles, David Chee Guan Foo, et al. 2014. Geographic factors are associated with increased risk for out-of-hospital cardiac arrests and provision of bystander cardio-pulmonary resuscitation in Singapore. *Resuscitation* 85, 9 (2014), 1153–1160.
 - [25] Dieter Oosterlinck, Dries F Benoit, Philippe Baecke, and Nico Van de Weghe. 2017. Bluetooth tracking of humans in an indoor environment: An application to shopping mall visits. *Applied geography* 78 (2017), 55–65.
 - [26] C Pommerenke, S Poloczek, F Breuer, J Wolff, and J Dahmen. 2023. Automated and app-based activation of first responders for prehospital cardiac arrest: an analysis of 16,500 activations of the KATRETTTER system in Berlin. *Scandinavian Journal of Trauma, Resuscitation and Emergency Medicine* 31, 1 (2023), 105.
 - [27] Antti Rasmus, Mathias Berglund, Mikko Honkala, Harri Valpola, and Tapani Raiko. 2015. Semi-supervised learning with ladder networks. *Advances in neural information processing systems* 28 (2015).
 - [28] David E Rumelhart, Geoffrey E Hinton, and Ronald J Williams. 1986. Learning internal representations by error propagation, parallel distributed processing, explorations in the microstructure of cognition, ed. de rumelhart and j. mcclelland. vol. 1. 1986. *Biometrika* 71, 599–607 (1986), 6.
 - [29] Manasvi Saha, Michael Saugstad, Hanuma Teja Maddali, Aileen Zeng, Ryan Holland, Steven Bower, Aditya Dash, Sage Chen, Anthony Li, Kotaro Hara, et al. 2019. Project sidewalk: A web-based crowdsourcing tool for collecting sidewalk accessibility data at scale. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems*. 1–14.
 - [30] A Siegel. 1975. The development of spatial representations of large-scale environments. *Advances in Child Development and Behavior/Academic Press* (1975).
 - [31] Christopher M Smith, Sarah N Lim Choi Keung, Mohammed O Khan, Theodoros N Arvanitis, Rachael Fothergill, Christopher Hartley-Sharp, Mark H Wilson, and Gavin D Perkins. 2017. Barriers and facilitators to public access defibrillation in out-of-hospital cardiac arrest: a systematic review. *European Heart Journal-Quality of Care and Clinical Outcomes* 3, 4 (2017), 264–273.
 - [32] Petros Spachos and Konstantinos N Plataniotis. 2020. BLE beacons for indoor positioning at an interactive IoT-based smart museum. *IEEE Systems Journal* 14, 3 (2020), 3483–3493.
 - [33] Yilin Sun, Yinan Dong, E Owen D. Waygood, Hamed Naseri, Yuhao Jiang, and Yijie Chen. 2023. Machine-learning approaches to identify travel modes using smartphone-assisted survey and map application programming interface. *Transportation research record* 2677, 2 (2023), 385–400.
 - [34] Wojciech Timler, Filip Jaskiewicz, Joanna Kempa, and Dariusz Timler. 2024. Automatic external defibrillator (AED) location—seconds that save lives. *Archives of Public Health* 82, 1 (2024), 153.
 - [35] Rui Wang, Fanglin Chen, Zhenyu Chen, Tianxing Li, Gabriella Harari, Stefanie Tignor, Xia Zhou, Dror Ben-Zeev, and Andrew T. Campbell. 2014. StudentLife: assessing mental health, academic performance and behavioral trends of college students using smartphones. In *Proceedings of the 2014 ACM International Joint Conference on Pervasive and Ubiquitous Computing* (Seattle, Washington) (*UbiComp '14*). Association for Computing Machinery, New York, NY, USA, 3–14. doi:10.1145/2632048.2632054
 - [36] Kasper Welbers, Elly A Konijn, Christian Burgers, Anna Bij De Vaate, Allison Eden, and Britta C Brugman. 2019. Gamification as a tool for engaging student learning: A field experiment with a gamified app. *E-learning and Digital Media* 16, 2 (2019), 92–109.
 - [37] Line Zinckernagel, Carolina Malta Hansen, Morten Hulvej Rod, Fredrik Folke, Christian Torp-Pedersen, and Tine Tjørnhøj-Thomsen. 2016. A qualitative study to identify barriers to deployment and student training in the use of automated external defibrillators in schools. *BMC Emergency Medicine* 17 (2016), 1–12.

A Appendices

A.1 In-App Survey Questions

- Q1: *Memory Reliance* assesses reliance on memory versus app guidance to locate AEDs (1: complete reliance on app guidance; 4: complete reliance on memory).
- Q2: *Perceived Ease* gauges users' perceptions of the session difficulty (1: Very Difficult; 5: Very Easy).
- Q3: *Pre-Visit Familiarity* measures prior familiarity with AED locations, essential for evaluating its educational impact (1: Totally Unaware; 5: Super Familiar).
- Q4: *Post-Visit Confidence* evaluates users' confidence in retrieving AEDs after using the application, indicating the effectiveness of AEDHunter in enhancing emergency preparedness (1: Not Confident at All; 5: Very Confident).
- Q5: *Post-Visit Assistance Willingness* assesses willingness to assist others in AED retrieval, reflecting AEDHunter's impact on fostering.

A.2 Case Studies of Identified Challenges

We integrated survey data and sensor data to identify patterns and understand user experiences. Post-survey feedback indicated that AED3 was perceived as the most challenging to locate. For example, as shown in Figure 4, a participant from the map group spent a considerable amount of time during the Indoor AED Search phase despite covering very short physical distances. In the post-survey, this user cited limitations of the GPS indoor navigation system as the reason for difficulty in locating AED3. Participants in the no-map group also encountered significant challenges in finding AED3. They reported confusion caused by the building's two entrances, which led them to initially choose incorrect paths and waste time. Additionally, the AED's placement on an inner wall facing inward made it difficult to spot from the exterior, causing users to overlook it initially and spend extra time searching deeper inside the building.

Furthermore, while prior visits to AED locations generally improved participants' ability to relocate them, challenges remained, such as navigating environments lacking external visual cues and the potential for reduced situational awareness when using map

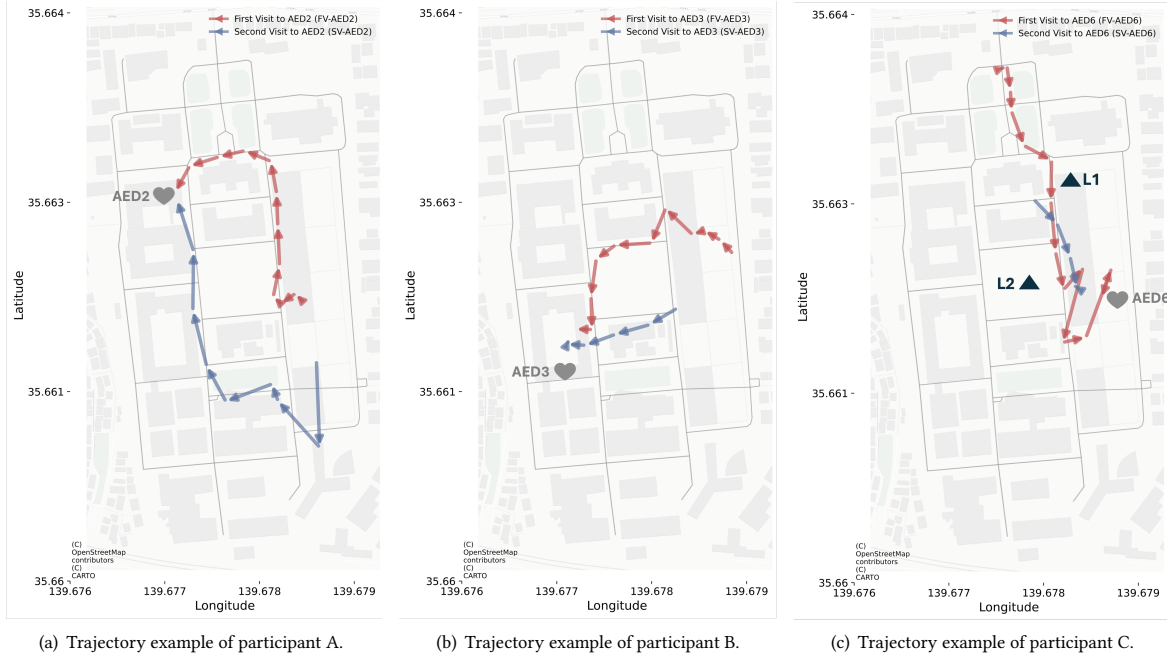


Figure 8: Examples of participants' trajectories visualized using GPS data. Participant B and C from Map group. Participant A from No Map group. L1 and L2 refer to landmark 1 and landmark2.

assistance. In indoor environments lacking external visual cues, participants could become disoriented. For example, in Figure 8(a) (SV-AED2), Participant A (without map assistance) began his trip on the second floor of a building. Despite having previously visited this AED (SV-AED1) and rating his pre-visit familiarity as high (4 out of 5), he initially chose a path that led away from the AED, resulting in detours and a longer route. While inside the building, he believed the AED was located to the southeast, but only upon exiting and observing the surrounding environment did he realize it was actually to the northwest. He then adjusted his route and

proceeded directly to the AED via the shortest path. Additionally, participants with map assistance may focus more on the map within the application and pay less attention to their surrounding environment. For instance, Participant B (with map assistance) reported feeling unfamiliar with the surroundings when starting her second visit to an AED from the middle of a previously traveled path (Figure 8(c)). Missing her initial reference point L1 (the side view of the building group) and not yet having reached the next landmark such as L2, she felt disoriented and required map assistance to navigate effectively.